

James Bai

Engineer @ Xero · 5+ yrs · .NET · AWS · React · LLM firefighter

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Wellington · NZ

/**

* Software engineer with 5+ years shipping .NET and AWS systems at Xero, currently on
* App Store billing and AI/MCP product infrastructure.
*
* I use AI heavily, but my value is knowing where to point it, how to verify the result,
* and how to repair the system when it gets things wrong.
* Claude Code and Codex are daily tools. My side project is a ~457,000 line AI powered
* ERP I built and operate alone, with ~10 daily users.
*
* @objective Senior software engineering roles
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experience

Xero · Engineer

2023.11 – present

stack: C# / .NET 8/10 · AWS Lambda · SQS · SNS · DynamoDB · AWS Glue · React / TypeScript · Postgres

sentinel/p0	Feature lead on two features for the global App Store usage-based billing rewrite. Designed and shipped the .NET / C# APIs and Postgres schema that became the billing core . Live ~1 month, only one minor bug found . Also built the AWS Glue aggregation pipeline.
ai/mcp-team	Selected to a new team owning Xero's official AI/MCP surface , including the Xero MCP server . Building applied AI infrastructure across LLM tool use , agent workflows, retrieval/context design, eval checks, observability, privacy, and safety guardrails.
plan-migration	Co-designed and shipped React / Refine / MUI SPA backed by a .NET BFF for App Store merchant self-service.
manual-invoicing	Feature lead on the manual-invoicing flow that recovered hundreds of thousands of dollars in overdue invoices. Leadership shout-out.
us-tax-support	On-call owner for the US Sales Tax service supporting Xero's US merchants: production incidents, schema fixes, customer-facing data corrections.
also	Deep Embed Service · CodeRed CI/CD · shared frontend libs · PCI compliance · API latency · associate mentoring.

Section6 · Software Consultant

2021.03 – 2023.11

stack: C# / .NET · Azure · React / TypeScript · Python · Kubernetes

virtual-cfo	Built .NET / C# APIs on Azure Functions integrating Xero & BambooHR, paired with React / TypeScript / Fluent UI dashboards for finance teams. APIs processed 66k+ records/year in seconds.
bnz-migration	Red Hat partnership. Python/Bash automation migrated 270 of 350 Java apps off legacy Kubernetes within a year.
vodafone-dx	Owned Payment, SMS, and Email gateways on .NET / C# · Azure · Kubernetes , serving Vodafone NZ's digital experience platform.
also	Multi-client consulting across banking, telco, and SaaS · CI/CD on TeamCity & GitHub Actions · code review and junior mentoring.

side-projects

BYDH ERP • AI powered, solo build

~457,000 LOC • ~10 daily users • 1 maintainer

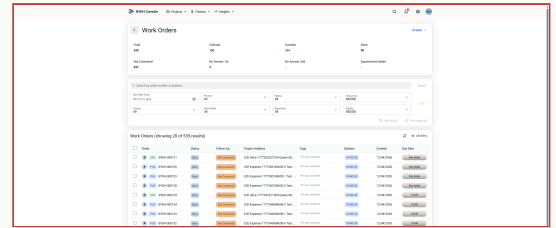
A custom ERP for a construction business. **Designed, built, and operated end to end** as an AI native system. Web + native mobile.

email→ledger Gemini parses inbound bills, classifies, and creates payable/receivable records. Review screens validate source rows before save; settlement **auto-creates the payment entry**.

ai-quoting Wired the system to MCP so ChatGPT can drive quoting via tool calls, structured context, pricing lookup, preview, and controlled save flow.

insights Cashflow, P&L, project profitability dashboards.

mobile React Native + Expo app for on site field use; shares auth + AWS Cognito with the web app.



stack: react • ts • .net • aws (cloudfront, rds, lambda) • postgres

→ case study at /bydh

Selected live web products

React • Tailwind • C# • AWS

Success Health

successhealth.co.nz

Built and operate clinic booking and customer-facing flows for a Wellington physio.

AHU Handyman

ahuhandyman.com.au

Public site for an Australian handyman business, focused on service discovery, lead capture, and simple production deployment.

ml-vision

pixel-style-transfer Neural style transfer in **PyTorch**: Gatys + AdaIN + Johnson, VGG-19 features, pixel-art grid-preservation loss, ControlNet baseline.

cv-self-training-counter CV auto-label (Dual-Otsu + SimpleBlobDetector) → Label Studio → **YOLOv8s** self-training loop → CoreML export.

receipt-ocr **Florence-2** VLM + **PaddleOCR** with line-grouping algorithm and AWS Lambda container deploy plan.

skills

lang C# / .NET • TypeScript • Python
web React • Redux • Next • MUI • Fluent • Next UI
mobile React Native • Expo
data Postgres • MySQL • DynamoDB • Dapper • EF • Flyway
msg SQS • SNS

cloud AWS (Lambda, SQS, SNS, DynamoDB, Glue, ECS, S3, RDS, **Cognito**, CloudFront) • Azure
platform K8s • Docker • Terraform • CloudFormation
ci/cd GitHub Actions • TeamCity • Jenkins
ai Claude Code • Codex • Gemini • MCP • LLM tool use • agents • RAG/context • evals

education + cert

PGDipSci (Merit), Computer Science

University of Auckland • 2020 — 2024

AWS Certified Developer Associate

Amazon Web Services • 2026

Azure Developer Associate • Microsoft

2020 — 2022

BSc, CS & Statistics (AI specialisation)

Victoria Univ. of Wellington • 2015 — 2018

RHCSA • Red Hat

2022 — present